KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026-2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST – I
(Regulations 2022)

Month and Year	July 2026	Roll Number	23CSL266
Programme	B.E/BTech	Date	06-07-2026
Branch	Common to all branches	Time	09.15am – 10.45am
Semester	VII	Duration	1½ Hours
Course Code	22GCT71	Max. Marks	50
Course Name	Engineering Economics and Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Compare micro and macro economics analysis with an example. (2) CO1 K2
2. Report four Basic assumptions in Economics. (2) CO1 K2
3. If the demand equation is given by D=1000-P, and the supply equation is S=100+4P, determine the price at equilibrium. (2) CO1 K3
4. If the price of milk increases, what do you think will happen to the demand of cornflakes? (2) CO1 K3
5. Outline the concept of discounting principle. (2) CO1 K2
6. Intrepret the concept of Market equilibrium with an example. (2) CO1 K2
7. Classify the methods for measuring national income. (2) CO2 K2
8. Calculate the GNP of a country, given GDP is Rs.34000 crores and net income from abroad is Rs.4200 crores. (2) CO2 K3
9. Compare demand pull inflation and cost pull inflation. (2) CO2 K3
10. Recall the various stages in Business Cycle (2) CO2 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Illustrate the circular flow of economic activities in two sector and four sector economy. (10) CO1 K2
12. A retail store decides to execute a planned price hike on its premium bicycle model at the start of the summer season. Immediately following the price increase, the store manager notices a sharp decline in the weekly volume of bicycles purchased by walk-in customers. Explain how this scenario demonstrates the core Law of Demand and state the typical relationship it illustrates between price and quantity. (10) CO1 K3
13. Interpret the causes of inflation and measures taken to control inflation. (10) CO2 K2
14. A nation experiences consecutive quarters of declining factory output, dropping sales for commercial trucks, and widespread worker layoffs. After reaching its absolute lowest point, retail demand slowly begins to rise again, prompting firms to slowly resume hiring.Identify the specific phases of the business cycle described in this sequence of economic events. (10) CO2 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	53.3	46.7	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - II
Regulation 2022

Month and Year	August 2026	Roll Number	23CSL267
Programme	BE/BTech	Date	17/08/2026
Branch	common to All branches	Time	9:15 AM - 10:45 AM
Semester	S7	Duration	1 Hour 30 Min
Course Code	22GCT71	Max. Marks	50
Course Name	Engineering Economics and Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Classify the primary forms of business ownership models applicable to engineering enterprises. (2) CO2 K2
2. Contrast any two F.W. Taylor's Scientific Management principles with Henri Fayol's Administrative Management concepts. (2) CO2 K2
3. Brief the essential managerial skills required across different levels of management. (2) CO2 K2
4. By what level of management, the organizational issue will be solved or handled in a project team? (2) CO2 K2
5. Outline the major stages involved in the New Product Development process. (2) CO3 K2
6. Explain how product sales and profit characteristics vary across different stages of the Product Life Cycle. (2) CO3 K2
7. Apply Cost-Plus Pricing and Value-Based Pricing to a newly engineered product and compare the selling prices obtained using both methods. (2) CO3 K3
8. Briefly narrate the significance of patent in protecting technological innovations. (2) CO3 K2
9. Identify the pricing strategy in which a high price is charged for a new technology product during its initial launch. (2) CO3 K2
10. Apply the Cost Plus Pricing method to determine the selling price when the cost of a product is ₹50 and the markup is 20%. (2) CO3 K3

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Explain the essential functions of management in detail. Illustrate how top management, middle management, and first-line supervisors fulfil their specific administrative and operational roles within a multi-divisional engineering company. (10) CO2 K2
12. An engineering startup company is developing a patented autonomous drone for agricultural crop monitoring. (10) CO3 K3
    Apply Henri Fayol's administrative management principles to design an efficient organizational structure that ensures effective authority and unity of command within the engineering team.
    Construct a tailored marketing mix strategy (4 Ps: Product, Price, Place, Promotion) suitable for successful market entry and adoption in agricultural sectors.
13. Describe the complete step-by-step pipeline of New Product Development (NPD) from concept generation and testing to full-scale commercialization. Illustrate how strategic pricing decisions (Price Skimming vs. Penetration Pricing) evolve as the newly engineered product moves through the four stages of its Product Life Cycle (PLC). (10) CO3 K2
14. The new wing of the TVS Motor corporation specializing in smart EV battery management systems is planning to expand its market reach. If you are in that team, by applying appropriate Intellectual Property Rights (IPR) strategies how will you safeguard the proprietary firmware and hardware design before entering international markets. And develop a comprehensive pricing strategy by utilizing Cost-Plus and Value-Based pricing models across different customer segments to maximize market penetration (10) CO3 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	60	40	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI, ERODE – 638 060
CONTINUOUS ASSESSMENT TEST – II - Retest
Regulations 2022

Month and Year : Oct 2026	Roll Number:
Programme : B.E/B.Tech	Date :
Branch :	Time :
Semester : VII	
Course Code : 22GCT71	Duration : 1 ½ Hours
Course Name : Engineering Economics and Management	Max. Marks : 50

PART - A (10 × 2 = 20 Marks)
ANSWER ALL THE QUESTIONS

1.	Differentiate between sole proprietorship and partnership based on any two characteristics.	[CO2,K2]
2.	Use Taylor's scientific management principles to improve operational efficiency.	[CO2,K3]
3.	Compute the functions to be carried for the effective management of any organisation.	[CO2,K3]
4.	Identify any four managerial skills essential for top-level management.	[CO2,K2]
5.	Interpret the differences among customer needs, wants, and demands with an example.	[CO3,K2]
6.	Show how a price-skimming strategy can be applied to a newly launched product with a suitable example.	[CO3,K3]
7.	Compare trademarks and copyrights based on their purposes and protections.	[CO3,K2]
8.	Draftthe characteristics of tangible property and intellectual property with suitable examples.	[CO3,K3]
9.	Mentionthe significance of a price-skimming strategy for a newly launched product	[CO3,K2]
10.	Highlight the importance of relationship marketing in retaining customers.	[CO3,K2]

Part – B (3 × 10 = 30 Marks)
ANSWER ANY THREE QUESTIONS

11.	Apply Taylor's scientific management principles and Fayol's administrative management principles and compare their approaches to managing an organisation.	(10)	[CO2,K3]
12.	Illustrate the characteristics of franchising and mergers with suitable examples.	(10)	[CO2,K3]
13.	Explain the various stages involved in the new product development process.	(10)	[CO3,K2]
14.	Describe the different stages of the product life cycle with examples.	(10)	[CO3,K2]

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	-	60%	40%	-	-	-

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - III
Regulation 2022

Month and Year	October 2026	Roll Number	23512bb
Programme	BE/BTech	Date	07.10.2026
Branch	Common to All branches	Time	09.15am -10.45am
Semester	S7	Duration	01 h 30 Min
Course Code	22GCT71	Max. Marks	50
Course Name	Engineering Economics and Management		

{{instructions}}

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Classify the different functions of production planning and control. (2) CO4 K2
2. A company is selecting a new plant location based on availability of raw materials, labour and transportation. Which site-selection factors should be considered? (2) CO4 K2
3. Identify the types of production system you will prefer for Pencil manufacturing and Bakery shop. (2) CO4 K2
4. Identify the parameters of various inventory management systems (2) CO4 K2
5. Describe how EOQ helps in inventory management. (2) CO4 K2
6. Outline the steps in the accounting cycle. (2) CO5 K2
7. A manufacturing company purchased an automatic machine for Rs. 2,19,500. estimated salvage value of Rs. 19,500 and an estimated working life of 4 years. Compute the depreciation charge for each of the 4 years. (2) CO5 K3
8. Infer the significance of Break-even analysis. (2) CO5 K2
9. An industry is planning to install a new Heat Recovery system which costs around 5 Lakhs. Annual cash inflows for the next 5 years are Rs.80,000. Calculate the payback period of the new project. (2) CO5 K3
10. Outline the importance of Net Present Value (NPV) method used in capital budgeting. (2) CO5 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Explain the type of layouts suitable for 1) hospitals and 2) automobile assembly unit with neat sketches. (10) CO4 K2
12. Krishna industry needs 24,000 units/year of a bought out component which will be used in its main product. The ordering cost is Rs. 150 per order and the carrying cost per unit per year is 18% of the purchase price per unit. The purchase price per unit is Rs.75. Find 1) EOQ, 2) No. of orders per year, 3) Time between successive orders and 4) Total inventory cost. (10) CO4 K3
13. Explain breifly any four concepts followed in financial accounting practices. (10) CO5 K2
14. A company makes a single product with a sales price of Rs.10 and a variable (13) cost of Rs.6 per unit. Fixed costs are Rs.60,000. Calculate i) Number of units to break-even ii) Sales at break-even iii) Contribution to sales ratio (in terms of percentages) iv) What number of units will need to sold to achieve a profit of Rs.10000? What level of sales will achieve a profit of Rs.30,000? (10) CO5 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	60	40	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026-2027(ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST – I
(Regulations 2022)

Month and Year	July 2026	Roll Number	23CSL266
Programme	B.E.	Date	06.07.2026
Branch	CSE	Time	02.45pm-04.15pm
Semester	VII	Duration	1½ Hours
Course Code	22CSC71	Max. Marks	50
Course Name	Deep Learning		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

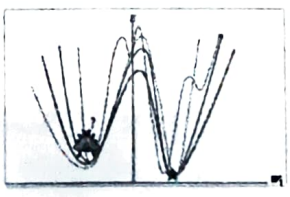
1. Classify the following physical quantities as either scalar or vector: (2) CO1 K3
   1.An athlete running a distance of 400 meters.
   2.A downward gravitational force of 9.8 Newtons is required to break the stone.
2. Transform the given image into a feature vector, which can be given as a input for a nueral network. (2) CO1 K3



3. Compare biological neurons and artificial neurons in terms of decision making, learning and computational efficiency. (2) CO1 K2
4. Give the mathematical expressions for Binary Cross-Entropy Loss and Categorical Cross-Entropy Loss. (2) CO1 K2
5. A neural network is being built to classify handwritten digits from the MNIST dataset. The model must first flatten each 28x28 image, then learn patterns using two hidden layers, and finally predict one of the 10 digit classes. Identify the missing keywords in the code for this classification (2) CO1 K3

```
model = _____([
    _____(input_shape=(28, 28)),
    _____(128, activation='relu'),
    _____(64, activation='relu'),
    _____(10, activation='softmax')
])
```

6. A hospital needs to predict the risk of diabetes using Blood Glucose Level (x_1), BMI (x_2), and Age (x_3) with the help of neuron. The neuron recieves input as x_1=0.85, x_2=0.65, x_3=0.40 with weights w_1=0.5, w_2=-0.8, w_3=-1.2, and bias b=0.4.Compute the output by applying ReLu activation. (2) CO1 K3
7. Differentiate between lasso regression and ridge regression. (2) CO1 K2
8. A deep neural network is trained using Gradient Descent on the non-convex error surface as shown in the Figure. The optimization process converges to a local minimum instead of the global minimum. Identify the best fit Gradient Descent type for the network. (2) CO2 K3



9. Outline the purpose of the method 'torch.load()' in PyTorch. (2) CO2 K2
10. Interpret the limitations of vanilla Mini-batch Gradient Descent in optimizing deep neural networks. (2) CO2 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. A campus placement cell employs a single-neuron neural network to predict whether a student is likely to be placed based on three normalized features: CGPA (x_1), Programming Skill Score (x_2), and Aptitude Test Score (x_3). The neuron uses the following weights and bias: (10) CO1 K3

w_1=0.5
w_2=-1.0
w_3=2.0
Bias, b=-2.0

(a) Calculate the net input (z) for each student.

(b) Determine the output of the neuron for each student using the following activation functions:

1.Sigmoid 2.ReLU

(c) Compare the outputs obtained using the two activation functions and identify which is best.

The placement data of five students are given below.

Student	CGPA (x_1)	Programming Skill (x_2)	Aptitude Score (x_3)
S1	2	1	3
S2	1	2	2
S3	3	2	1
S4	2	3	2
S5	1	1	1

12. A financial institution develops a fully connected feed-forward Multi-Layer Perceptron (MLP) to predict whether a loan application should be approved based on 10 customer attributes (such as age, income, credit score, employment status, and loan amount). The network consists of an input layer with 10 neurons, two hidden layers containing 4 and 3 neurons, respectively, and an output layer with one neuron. Assume that bias not assigned. (10) CO1 K3

(a) Construct the neural network and label the number of neurons and the connections between successive layers.

(b) Determine the total number of trainable weight parameters between each pair of consecutive layers in the network.

13. Illustrate the role of differentiation in Gradient Descent for minimizing the loss function and compare the working principles of Batch, Stochastic, and Mini-batch Gradient Descent. (10) CO1 K2

14. Describe the concept of model identifiability and explain how the equivalent network configurations produce identical outputs for the same input. (10) CO2 K2

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	50	50	0	0	0

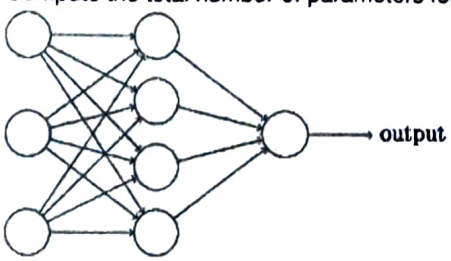
KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - II
Regulation 2022

Month and Year	August 2026	Roll Number	23CSL266
Programme	BE	Date	17/08/2026
Branch	CSE	Time	2:45 PM - 4:15 PM
Semester	S7	Duration	1.30hrs
Course Code	22CSC71	Max. Marks	50
Course Name	Deep Learning		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Infer the role of the parameter in AdaGrad. (2) CO2 K2
2. Why does an ill-conditioned Hessian matrix cause difficulties in the optimization process? (2) CO2 K2
3. Compute the total number of parameters for the given neural network. (2) CO2 K3



4. Identify the advantages of using Momentum-based optimization over plain gradient descent? (2) CO2 K2
5. Give the role of padding in convolutional layers, and how does it impact the output size? (2) CO3 K2
6. Provide a simple example to illustrate the working of Max pooling in CNN. (2) CO3 K2
7. An input image has dimensions 12 × 12. If a convolution layer uses 3 × 3 filters with 20 kernels, determine the number of activation maps produced at the output. (2) CO3 K3
8. Is it possible to use CNN to replicate artistic styles. If yes, give justification. (2) CO3 K2
9. List any two applications of CNNs in computer vision. (2) CO3 K2
10. Consider a CNN layer with the following parameters, Number of filters are 32 and Filter size is 2x2. Calculate the number of Parameters for a black and white image. (2) CO3 K3

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Demonstrate the working of any two adaptive learning rate algorithm. (10) CO2 K2
12. Consider the CNN architecture below with two convolutional layers followed by max-pooling and two fully connected (dense) layers. Calculate the number of parameters and output sizes for each layer. (10) CO3 K3

1. Convolutional Layer 1:
   - Input Image: 32x32x3
   - Number of Filters: 16
   - Filter Size: 3x3
   - Padding: 1

2. Max-Pooling Layer 1:
   - Pooling Size: 2x2
   - Stride: 2

3. Convolutional Layer 2:
   - Number of Filters: 32
   - Filter Size: 3x3
   - Padding: 1

4. Max-Pooling Layer 2:
   - Pooling Size: 2x2
   - Stride: 2

5. Flatten layer

6. Fully Connected Layer 1:
   - Number of Neurons: 128

7. Fully Connected Layer 2 (Output Layer): 10 Neurons

13. A digital photography application transforms ordinary photographs into different artistic styles such as sketches, painting and texture effects. Apply the concept of convolutional filters to explain how different visual features can be extracted and emphasized to create these artistic effects. (10) CO3 K3

14. Write a Python program to build, train, and evaluate a CNN model for classifying fruit images. Include suitable convolution and pooling layers and display the model summary and test accuracy. (10) CO3 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	40	60	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - III
Regulation 2022

Month and Year	October 2026	Roll Number	23CSL266
Programme	BE	Date	07/10/2026
Branch	B.E CSE	Time	2:45 PM - 4:15 PM
Semester	S7	Duration	1 Hour 30 Min
Course Code	22CSC71	Max. Marks	50
Course Name	Deep Learning		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. State the purpose of embedding in representation learning. (2) CO4 K2
2. A trained autoencoder produces a high reconstruction error for its input. What does this indicate about the quality of the learned representation? (2) CO4 K3
3. A text corpus contains the words "The student reads books". Find how word embeddings can represent these words as numerical vectors. (2) CO4 K3
4. Compare between RNN and LSTM. (2) CO4 K2
5. Illustrate the role of the hidden state in an RNN. (2) CO4 K2
6. A CNN shows high training accuracy but poor testing accuracy. Identify the likely problem and suggest one suitable technique to reduce it. (2) CO5 K3
7. A sentiment model fails to retain important words occurring earlier in a long review. Identify how the LSTM cell state can help retain such information. (2) CO5 K3
8. A network intrusion dataset contains normal traffic and several attack categories. Suggest a suitable output-layer configuration for multi-class intrusion classification. (2) CO5 K3
9. A speech dataset contains recordings with different durations. Identify one method by which the input sequences can be prepared for training a deep learning model. (2) CO5 K3
10. A CNN model needs to reduce the spatial dimensions of its feature maps while retaining important features. Identify a suitable layer and explain its function. (2) CO5 K3

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Draw and explain the architecture and working of a Recurrent Neural Network (RNN). Describe the role of hidden states in processing sequential data. (10) CO4 K2
12. . (10)
    a) Summarize the approaches used for sequence analysis. Discuss the role of Neural N-Grams in modelling sequential data. (5) CO4 K2
    b) Describe the working of a Part-of-Speech tagging system. Describe how sequence models can be used to assign POS tags to words in a sentence. (5) CO4 K2
13. Write a Python program to implement an autoencoder for dimensionality reduction. Use a suitable dataset, construct the encoder and decoder, train the model, and extract the latent representation of the input data. (10) CO5 K3
14. A university wants to develop a voice-based system that converts spoken commands into text. The speech recordings differ in duration, pronunciation, and speaking speed. Explain how a deep learning-based speech recognition system can be developed for this application. Discuss audio preprocessing, feature extraction, sequence modelling, handling variable-length speech, and output generation. (10) CO5 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	43.3	56.7	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026-2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST – I
(Regulations 2022)

Month and Year	July 2026	Roll Number	
Programme	B.E./BTech	Date	07-07-2026
Branch	Common	Time	09.15am – 10.45am
Semester	VII	Duration	1½ Hours
Course Code	22GEE02	Max. Marks	50
Course Name	Total Quality Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Based on the quality (Q) value, recommend the parameter that should be improved when Q<1. (2) CO1 K3
2. Show any four dimensions of quality in TQM. (2) CO1 K2
3. Classify the types of quality costs. (2) CO1 K2
4. Identify four obstacles associated with TQM implementation. (2) CO1 K2
5. Interpret the six basic TQM concepts, which were the most effective in World war-II. (2) CO1 K2
6. Give the difference between quality control and quality assurance in TQM. (2) CO1 K2
7. Show how does employee satisfaction relate to customer satisfaction. (2) CO2 K2
8. Using the American Society for Quality (ASQ) survey findings, suggest the factors that should be considered to influence an end user's purchase decision. (2) CO2 K3
9. Draw the hierarchy diagram for Maslow's Hierarchy of needs. (2) CO2 K2
10. Outline the advantages of an empowered team. (2) CO2 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Discuss the historical review of quality control in Total Quality Management. (10) CO1 K2
12. Illustrate the 7 habits of highly effective people to bring harmony in work place. (10) CO1 K2
13. Make use of Deming's fourteen principles for quality management and relete how these principles can be effectively applied to improve quality in modern industry. (10) CO1 K3
14. Electronics products such as mobile, smart TV, etc., are defected easily. These products are often serviced by the service centre. How can a dealer use the service department to encourage the future electronic products sales? (10) CO2 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	60	40	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - II
Regulation 2022

Month and Year	August 2026	Roll Number	23CSL266
Programme	BE	Date	18/08/2026
Branch	Mechanical	Time	9:15 AM - 10:45 AM
Semester	S7	Duration	1 Hour 30 Min
Course Code	22GEE02	Max. Marks	50
Course Name	Total Quality Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Identify the ways to improve a process. (2) CO2 K3
2. Interpret the Juran's Triology. (2) CO2 K2
3. Outline the key elements of supplier partnership in a quality environment. (2) CO2 K2
4. Show the elements in 5S with suitable example. (2) CO2 K2
5. Develop a simple flowchart to prepare a coffee. (2) CO3 K3
6. Outline the steps involved in constructing a Pareto Diagram. (2) CO3 K2
7. Differentiate between attribute and variable. (2) CO3 K2
8. Schematically show the difference between positively skewed and negetively skewed distribution curves. (2) CO3 K2
9. Compare within piece variation and piece-to-piece variation. (2) CO3 K2
10. Determine Cp and Cpk for a process with an average 8.25, $\sigma = 0.04$, having USL = 8.35, and LSL = 8.2. (2) CO3 K3

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. A restaurant has long waiting time due to high rating in terms of quality of food. Illustrate PDSA cycle along with the seven phases to reduce the waiting time of customers. (10) CO2 K2
12. The following table gives the sample means and ranges for **10 samples of size 6** obtained from the production of a certain component. (Take A_2=0.483, D_3=0, and D_4=2.004) (10) CO3 K3

Sample No.	1	2	3	4	5	6	7	8	9	10
Mean (\bar X)	37.5	49.8	51.5	59.2	54.7	34.7	51.4	61.4	70.7	75.3
Range (R)	9.5	12.8	10.0	9.1	7.8	5.8	14.5	2.8	3.7	8.0

Construct X chart, R chart and comment the state of the process.

13. A painter paints the building but soon the paint fades due to several reasons. Identify the issues using cause and effect diagram and suggest necessary steps to prevent. (10) CO3 K3
14. Explain the Process Capability in detail and its significance. How is it different from Control Chart? (10) CO3 K2

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	56.7	43.3	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - III
Regulation 2022

Month and Year	October 2026	Roll Number	23CSL266
Programme	BE / BTech	Date	08.10.2026
Branch	CSE,IT, Mechanical	Time	09.15am-10.45am
Semester	S7	Duration	01 hour 30 min
Course Code	22GEE02	Max. Marks	50
Course Name	Total Quality Management		

{{instructions}}

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Infer any two purposes of the Quality tool 'Process Decision Program Chart'. (2) CO4 K2
2. Extend the significant uses of benchmarking. (2) CO4 K2
3. Considering the process specifications as 10 ± 3 for a particular quality characteristic and the average repair cost is $230, determine the loss fuction. Determine the loss at y = 12. (2) CO4 K2
4. Recommend the benefits of total productive maintenance. (2) CO4 K2
5. Show the stages of FEMA in quality management. (2) CO4 K2
6. Outline the benefits of adapting ISO quality systems. (2) CO5 K2
7. Outline the key elements of ISO 9001 : 2015. (2) CO5 K2
8. Classify the types of quality audits. (2) CO5 K2
9. Select suitable ISO standard for environmental management and justify your answer. (2) CO5 K2
10. Show any two barriers you will experience as a CTO of a software company when you want to implement a new quality strategy in your organisation. (2) CO5 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Construct the Quality Function Deployment (QFD) using the House of Quality (HoQ) for an automated laundry service to be installed in a college hostel. Identify the Voice of the Customer (customer requirements), technical requirements, importance ratings, relationship matrix, technical correlation, and overall priority of technical requirements. (10) CO4 K3
12. Develop a Failure Mode and Effects Analysis (FMEA) worksheet for an industrial conveyor belt system used for material handling. Identify at least five potential failure modes, their causes and effects, existing controls, and assign appropriate Severity (S), Occurrence (O), and Detection (D) ratings. Calculate the Risk Priority Number (RPN) for each failure mode, prioritize the risks, and recommend suitable corrective actions for the high-risk failure modes. (10) CO4 K3
13. Interpret how do organizational evaluation standards classified. Examine the requirements and benefits of ISO 14000. (10) CO5 K2
14. Choose any one industry and explain the process of implementing ISO. (10) CO5 K3

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	50	50	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026-2027(ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST – I
(Regulations 2022)

Month and Year	July 2026	Roll Number	23CS426b
Programme	B.E.	Date	07.07.2026
Branch	CSE	Time	02.45pm-04.15pm
Semester	VII	Duration	1½ Hours
Course Code	22CSE29	Max. Marks	50
Course Name	Software Project Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. List the components of a Business case. (2) CO1 K2
2. Write the objectives and sub objectives of Employee payroll project. (2) CO1 K2
3. Identify any two risks in the online food delivery project. (2) CO1 K3
4. Calculate the ROI of the project for which netprofit is Rs.40000 in 5 years and total investment is Rs.1,50,000. (2) CO1 K3
5. Give the purpose of programme brief in Strategic Programme Management. (2) CO1 K2
6. Draw the Gantt chart of any project of your interest. (2) CO2 K2
7. Mention the challenges in scope management. (2) CO2 K2
8. Identify and list the project deliverables of a project. (2) CO2 K2
9. List the items present in Product descriptions. (2) CO2 K2
10. Give the use of activity diagram in project planning. (2) CO2 K2

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. Calculate the net profit, payback period, return on investment and net present value for the given projects A and B using 5% and 8% discount rates commonly for both the projects.Analyze the results and decide the best project. (10) CO1 K3

Year	Project A	Project B
0	-1,00,000	-1,50,000
1	20,000	15,000
2	23,000	12,000
3	25,000	20,000
4	36,000	50,000
5	30,000	50,000

12. A college plans to develop a Student Management System (SMS) to automate academic and administrative activities such as student admission, course registration, attendance tracking, examination management, fee payment, and result processing. For this system, (10) CO1 K3
    Identify the key stakeholders
    Determine the objectives of developing the Student Management System.
    Illustrate the Project Management Life Cycle
    Explain how each phase of the project management life cycle would be applied to this project.
13. Illustrate the product breakdown structure and work breakdown structure with a suitable example. (10) CO2 K2
14. Outline the stepwise project planning with a neat flow diagram. (10) CO2 K2

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	60	40	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - II
Regulation 2022

Month and Year	August 2026	Roll Number	23CSL266
Programme	BE	Date	18/08/2026
Branch	CSE	Time	2:45 PM - 4:15 PM
Semester	S7	Duration	1.30hrs
Course Code	22CSE29	Max. Marks	50
Course Name	Software Project Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Identify the problems in over and under estimation of a project. (2) CO2 K2
2. List the software estimation techniques. (2) CO2 K2
3. Calculate the function points for a project that has 5 input types, 2 entity types referenced, and 2 output types using the unadjusted Mark II function points method. Use industry-average weights: input = 0.58, entity type referenced = 1.66, output = 0.26. (2) CO2 K3
4. For an embedded software project of size 20 KLOC, the sum of the COCOMO II scale factors is 25. Using A = 2.5, estimate the nominal effort in person-months using the COCOMO II Early Design model. (2) CO2 K3
5. Distinguish between top down approach and bottom up approach of software cost estimation. (2) CO2 K2
6. What is meant by a critical path in project management?. (2) CO3 K2
7. Bring out the objectives of activity planning. (2) CO3 K2
8. An activity has Earliest Start = Day 5, Earliest Finish = Day 8, and Latest Finish = Day 10. Compute the total float of the activity. (2) CO3 K3
9. An activity has an Optimistic time of 5 days, Most Likely time of 8 days, and Pessimistic time of 14 days. Determine its expected duration using the PERT formula. (2) CO3 K3
10. A project's critical path length is 20 days while the customer-imposed deadline is 25 days. Determine the total scheduling slack available and state whether crashing is necessary. (2) CO3 K3

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. The following data are available for previously developed software projects P, Q, R, S, T, U, V and W. Calculate the Symons Mark II Function Points for each project. Using the results, determine the relationship between Function Points and Developer Days, and estimate the effort required for the two new projects M and N. (10) CO2 K3

Project	Inputs	Outputs	Entity Access	System Users	Prog. Lang.	Developer Days
P	320	180	30	10	X	25
Q	480	1250	115	22	X	78
R	550	1350	80	20	Y	112
S	680	2200	100	190	Y	165
T	175	350	32	12	Z	24
U	260	900	60	28	Z	48
V	1500	1650	225	30	Y	310
W	600	820	120	8	Z	65
M	200	400	42	25	Y	—
N	520	1180	72	35	Y	—

12. Before starting a software development project, the team wants to predict the required effort and resources. Explain the various software effort estimation techniques and demonstrate their application using suitable examples. (10) CO2 K2

13. The following table gives the activities, their predecessors, and durations of a software project. Construct the Activity-on-Arrow network and identify the critical path and total project duration. (10) CO3 K3

Activity	Depend on	Duration(Days)
A	-	6
B	A	5
C	A	8
D	B	6
E	B	9
F	C	7
G	D,F	5
H	E	8
I	G,H	4
J	G	6
K	I,J	5

14. A software project encounters unclear user requirements, dependency on an unreliable external supplier, and lack of expertise in a newly introduced technology. Identify the appropriate risk categories and illustrate the risk identification, assessment, mitigation, and monitoring activities for each risk. (10) CO3 K2

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	50	50	0	0	0

KONGU ENGINEERING COLLEGE, PERUNDURAI – 638 060
(Autonomous)
2026 - 2027 (ODD SEMESTER)
CONTINUOUS ASSESSMENT TEST - III
Regulation 2022

Month and Year	October 2026	Roll Number	23CSL206
Programme	BE	Date	08/10/2026
Branch	B.E CSE	Time	2:45 PM - 4:15 PM
Semester	S7	Duration	1 Hour 30 Min
Course Code	22CSE29	Max. Marks	50
Course Name	Software Project Management		

PART - A (10 X 2 = 20 MARKS)

Answer all the questions

1. Four activities are reviewed at a progress meeting: Design is complete and on schedule; Coding is 3 days late but can be recovered by the team using overtime; Testing is 2 weeks late and can be recovered only by extra budget and a revised delivery date; Documentation is complete and 1 day early. Apply the RAG method to rate each activity. (2) CO4 K3
2. Distinguish between a Gantt chart and a slip chart as techniques for visualizing the progress of a project. (2) CO4 K2
3. A college wants a library management system. A ready-made software package is available that meets most of its needs, but it must be modified to match the college's book-issue rules. Identify whether the college should adopt a bespoke, off-the-shelf or COTS package and give a reason. (2) CO4 K3
4. Two supplier bids are scored out of 10. Supplier A scores 8 for cost and 5 for quality, and Supplier B scores 6 for cost and 9 for quality. The customer considers cost is 60% important and quality is 40% important. Decide which supplier should be selected and support your decision with the scores. (2) CO4 K3
5. Describe the role of the configuration librarian in controlling changes to a project's requirements. (2) CO4 K2
6. A newly formed software team is arguing over task allocation and members are challenging the project leader's decisions. Apply Tuckman's model of team development to identify the stage the team is in and state the stage that follows it. (2) CO5 K3
7. Outline the steps of the Delphi approach and infer how it reduces the disadvantages of group decision making. (2) CO5 K2
8. A developer believes that working harder will improve his performance and wants the promised bonus very much. However, he feels that the management will not give the bonus even if he performs well. Apply the expectancy theory of motivation to determine his likely level of motivation and justify your answer. (2) CO5 K3
9. Apply the communication genre matrix (same time / different time and same place / different place) to categorize the following modes of communication: daily stand-up meeting, e-mail, instant messaging, notice board and voicemail. (2) CO5 K3
10. A software project team is spread across three cities in different time zones. Apply the ideas of dispersed and virtual teams to identify two communication problems that may arise and suggest one measure to overcome each. (2) CO5 K3

PART - B (3 X 10 = 30 MARKS)

Answer any three questions

11. A project has a total budget of Rs.8,00,000 and is planned to be completed in 24 months. After 4 months, the project is 12% complete and Rs.1,20,000 has been spent. According to the plan, 20% of the work should have been completed by this time. Calculate the Cost Performance Index (CPI) and the Schedule Performance Index (SPI). Explain what these values tell you about the project's progress. (10) CO4 K3
12. A private engineering college has decided to buy a new student management system from an outside supplier. Using this situation as an example, explain the stages of contract placement. Describe what happens at each stage and why it is important in choosing a suitable supplier. (10) CO4 K2
13. Describe the different types of organizational structures and team structures used in organizations. Explain how each one supports smooth working and good teamwork. Give examples to show how they are used in practice. (10) CO5 K2
14. Explain the recruitment process for choosing the right candidate for a job. Describe the main steps, from identifying job requirements to final selection. Show how each step helps to find a suitable person and makes the organization more effective. Give examples where needed. (10) CO5 K2

Bloom's Taxonomy Level	Remembering (K1)	Understanding (K2)	Applying (K3)	Analysing (K4)	Evaluating (K5)	Creating (K6)
Percentage	0	60	40	0	0	0